

VOLUME
3

EMERGENCY SUB PLANS

for

MIDDLE SCHOOL MATH

KEEP LEARNING ON TRACK WHEN PLANS CHANGE

GRADES
6-8

NO-PREP
ACTIVITIES

READY-TO-USE
RESOURCES

PERFECT FOR
EMERGENCY
LESSONS

GREAT FOR
SUBSTITUTE
TEACHERS

PRINT OR
DIGITAL USE

RATIO & PROPORTIONS

Solve each problem.

1. The ratio of boys to girls in a class is 3:5. If there are 24 girls, how many boys are in the class?



2. Simplify the ratio 18:24.

GEOMETRY

Find the value of x .

 $x =$

SOLVE FOR x

Solve each equation.

1. $2x + 5 = 17$

2. $3(x - 4) = 21$

3. $\frac{x}{4} + 3 = 8$

4. $5x - 7 = 28$

PLOT & ANALYZE

The graph shows the number of skateboards sold each week.



1. How many skateboards were sold in Week 4?
2. Between which weeks did sales decrease?

ORDER OF OPERATIONS

Evaluate each expression.

1. $12 + 6 \times 2 - 4$
2. $(18 - 6) \div 3 + 7$
3. $2^4 + 3 \times (8 - 5)$
4. $24 \div (2 + 2) + 5$


MATH
EMERGENCY
KIT

YOUR CLASSROOM LIFESAVER


ENGAGING
AND EFFECTIVE

SAVES YOU
TIME

STUDENTS STAY
ON TRACK

MADE BY
TEACHERS
for
TEACHERS

Set this up while you are well

You are reading this because you are **well**. Spend fifteen minutes now and this pack will run itself on a morning when you are not.

The one job you have today

Print pages 3–6 and leave them in a clearly labeled folder where a substitute will find them — top drawer, on the desk, or at the front office. Everything a sub needs is in those four pages.

You do not need to print the lessons ahead of time. The sub picks one on the day and copies it.

SETTING IT UP • FIFTEEN MINUTES, ONCE

- 1 **Print the whole pack once.** Put it in a folder or binder. Write your name and room number on the cover.
- 2 **Print pages 3–6 a second time** and clip them to the front. Those are the pages a substitute reads first.
- 3 **Tell your department chair where it lives.** On a real sick day you will not be answering the phone, so somebody else needs to know.
- 4 **Cross off lessons as you use them.** There is a tracker on page 4. Two years of absences will never repeat if you keep it current.
- 5 **Optional: leave a class roster** and a note about students who need a reader, extra time, or specific seating.

Why nothing here needs technology

On a sub day the Chromebook cart is often locked, the login server is down, or nobody knows the room code. Every lesson runs on paper and a pencil.

Calculators are optional throughout, and each lesson says where one actually helps.

Why it is not busy work

Every lesson targets a real Grade 6–8 standard and produces graded evidence you can put in a folder.

Away for a week? Running four of these in sequence is a defensible unit, not a holding pattern.

About the answer keys

Every key sits on the **last page of its lesson** and shows the **working**, not just the answer. That is what lets a sub who has not taught math in years respond when a student asks “*but why?*”

If you also own the other volumes

Volume 1, Volume 2, Volume 4 contain thirty further lessons and not one of them repeats a topic from this pack. Keep the folders side by side and a substitute can cover weeks without a student seeing the same sheet twice.

You do not need to be a math teacher

You have walked into a middle school math class. You may not have done algebra since high school. That is completely fine — this pack was written for you.

Read this and you are ready

You do not need to be able to do the math. Every lesson is a paper worksheet with a complete answer key that shows the working. Your job is to hand it out, keep the room settled, and check answers against the key. You are not expected to teach the content.

DO THESE FOUR THINGS

- 1 **Pick one lesson** from the table on the next page. Any of them works for Grades 6, 7 or 8. If you cannot decide, use Lesson 01.
- 2 **Copy the student pages only.** Each lesson tells you exactly which page numbers. One per student.
- 3 **Keep the teacher page and the answer key.** Do not copy those.
- 4 **Read the teacher page once** before the bell. Two minutes, and it includes the exact words you can use to start.

If a student says “we already did this”

Ask them to prove it by finishing the challenge section on the last page. If they genuinely have, switch them to a different lesson — there are ten, and they will not have done all of them.

If they finish early

Every lesson has a challenge task on its final student page. There is also a discussion question on the teacher page you can run with the whole class for the last five minutes.

What to leave behind

Collect the sheets. Fill in the handover note on the last page of this pack — ninety seconds, and it tells the regular teacher which lesson you used, how far the class got, and anything that happened.

Leave both on the desk.

Choose your lesson

PICK ANY ONE

10 OPTIONS

Every lesson is standalone. There is no order and no prerequisite — pick whichever suits the class in front of you.

#	LESSON	WHAT IT COVERS	DIFFICULTY	MINS	STUDENT PAGES
01	Close Enough	Rounding & estimation	★	50	7–9
02	Point of Order	Decimal $\times$ and $\div$	★★	55	12–14
03	Change of Units	Measurement & conversion	★★	55	17–19
04	Both Directions	Absolute value & opposites	★	50	22–24
05	Balancing Act	Variables on both sides	★★★	55	27–29
06	Round Numbers	Circumference & area of circles	★★	55	32–34
07	Slide, Flip, Turn	Transformations	★★	60	37–39
08	Ask the Right People	Sampling, bias & inference	★	55	42–44
09	What Comes Next	Sequences & patterns	★★	55	47–49
10	Two Ways to Count	Two-way tables & frequency	★★★	55	52–54

Easiest to supervise

01, 04 and 06. Concrete and visual. A wrong answer usually looks wrong, so students catch themselves.

Best for a quiet class

02, 05 and 09. Steady practice with definite answers. Fast finishers will race — that is what the challenges are for.

Best for a restless class

03, 07 and 08. Kitchen conversions, cut-out shapes and survey design. Noisier, but productively so.

Used-lesson tracker • for the regular teacher

LESSON	DATE USED	CLASS	NOTES
01 — Rounding & estimation			
02 — Decimal $\times$ and $\div$			
03 — Unit conversion			
04 — Absolute value			
05 — Both-side equations			
06 — Circles			
07 — Transformations			
08 — Sampling & bias			
09 — Sequences			
10 — Two-way tables			

The first ten minutes

The first four minutes decide the next fifty. Here is a script that works even if you have never met this class.

SAY THIS AT THE DOOR

“Come in and take a seat. Bags off the desks. You will not need a Chromebook today — everything is on paper.”

THEN, ONCE THEY ARE SEATED

“Your teacher is out today. My name is _____. This is a real math lesson, not a free period. There is a worksheet, it gets collected at the end, and your teacher will see it.”

Why that last sentence matters

The single most useful thing you can say is that the work will be seen by their own teacher. It is also true — there is a handover note at the back of this pack.

THE SHAPE OF EVERY LESSON IN THIS PACK

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Settle, hand out sheets	Use the script above
5–10	Read the first panel together	Have a student read the worked example aloud
10–40	Students work through the tasks	Circulate. Check against the answer key
40–50	Challenge task for early finishers	Point them to the last student page
50–55	Collect sheets, pack up	Fill in the handover note

“I don’t get it”

Almost always means *I don’t know how to start*. Point at the worked example on the first page and ask them to copy its first line with their own numbers.

That single move unsticks most students without you needing to explain anything.

“Can I use a calculator?”

Unless the teacher page says otherwise, yes for the arithmetic and no for the reasoning tasks. Each teacher page states which.

If in doubt, allow it. A student doing the thinking with a calculator is better than one doing nothing.

Copying reminder

Student pages only. Each teacher page gives the exact range. The answer key is always the last page of the lesson — keep it.

Close Enough

2 MINUTES

Students round whole numbers and decimals to a stated place, estimate answers before calculating, then use estimation to judge whether a calculator answer is even possible. It ends with a supermarket trolley and the question every shopper actually asks: will fifty dollars cover it?

You do not need to know this subject

Every answer is a rounded number and the key prints all of them. There is no method to remember beyond looking at the digit to the right.

Photocopy these pages only

Pages 7–9

One per student. Do not photocopy this page or the answer key.

What students need

A pencil. Deliberately no calculators — the whole lesson is about not needing one.

WHAT TO SAY TO START

“Nobody works out a supermarket bill exactly while they are still in the aisle. They round, they add, and they know before they reach the till whether they have enough. Today you get good at that, and you find out why it is a real skill and not a lazy one.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–12	Read the rounding panel	Look at the digit to the right, and only that one
12–24	Task 1.1 — round and estimate	Steady, confidence-building
24–34	Task 1.2 — the trolley	Round to the nearest dollar
34–44	Task 1.3 — is it reasonable?	No calculating, just judging
44–50	Challenge, collect sheets	Discussion question below

Questions students will ask

“Which digit do I look at?”

The one immediately to the right of the place you are rounding to. Five or more rounds up; four or less stays. Nothing further right matters at all.

“Is estimating just guessing?”

No. A guess has no method. An estimate rounds first, then calculates exactly with the rounded numbers, so you can say how far out you might be.

“My estimate was not the exact answer. Is it wrong?”

No — that is the point of an estimate. It is wrong only if it is so far out that it would not have caught a real mistake.

If they finish early

Task 1.5 asks for the smallest and largest whole number that rounds to 400. Most students get the small end right and the large end wrong by one.

Curriculum focus • for the regular teacher’s records

Grade 6–7: rounding whole numbers and decimals to a given place value, and using estimation strategies to assess the reasonableness of answers. Discussion question for the last five minutes: “When is an estimate more useful than an exact answer?”

Rounding, and why it works

UNDERSTAND

16 MIN

NAME _____

CLASS _____

DATE _____

Look right, once

ROUNDING 3,462

to the nearest **10** 3,460
to the nearest **100** 3,500
to the nearest **1000** 3,000

Each time, look at the
next digit right.

5 OR MORE → ROUND UP
4 OR LESS → STAY

3,462 to the hundred:
the 6 says up → 3,500

3,462 to the thousand:
the 4 says stay → 3,000

ESTIMATING

Round **first**, then work
it out exactly.

$38 + 51$
 $\approx 40 + 50$
 $= 90$ (really 89)

Task 1.1 • Round it

DO

#	NUMBER	ROUND TO THE NEAREST	ANSWER
a	68	ten	
b	342	hundred	
c	1,650	hundred	
d	4,499	thousand	
e	8.76	tenth	
f	0.0847	hundredth	
g	12.5	whole number	
h	249.5	ten	

Task 1.2 • Estimate first

DO

Round each number, then work out the estimate. Do not calculate the exact answer.

#	CALCULATION	ROUNDED TO	ESTIMATE	#	CALCULATION	ROUNDED TO	ESTIMATE
a	$38 + 51$			d	6.9×8.1		
b	$412 - 187$			e	$587 \div 29$		
c	19×21			f	3.2×48		

Shopping and sanity checks

KEY SKILL

18 MIN

Task 1.3 • The trolley

APPLY

You have \$50. Round each price to the nearest dollar, add up your estimate, and decide before you reach the till.

ITEM	PRICE	ROUNDED
Bread	\$4.85	
Chicken	\$12.30	
Milk	\$2.99	
Cheese	\$7.45	
Detergent	\$18.60	
Apples	\$3.20	

a. Estimated total? _____

b. Will \$50 cover it? _____

c. Now add the real prices exactly.

d. How far out was your estimate? _____

e. Would rounding every price up have been safer? Why?

Task 1.4 • Is that even possible?

JUDGE

Somebody typed each of these into a calculator. Without working any of them out, decide whether the answer could be right. Write ok or no, and say roughly what it should be.

#	WHAT THE CALCULATOR SHOWED	OK / NO	ROUGHLY WHAT?
a	$4.2 \times 19 = 79.8$		
b	$312 \div 6 = 5.2$		
c	$0.5 \times 88 = 176$		
d	$9.8 + 10.4 = 20.2$		
e	$1,000 - 249 = 751$		
f	$24 \times 0.9 = 216$		

Task 1.4b • Round to a sensible place

THINK

There is no rule here. Decide what the situation needs and say why.

a. A crowd of 43,817 people at a stadium, reported on the news.

b. A patient's dose of medicine, 2.375 mL.

c. The distance from Earth to the Sun, 149,597,871 km.

d. The price of one screw when 200 cost \$37.

Errors and edges

STRETCH

16 MIN

Task 1.5 • Find the error

DETECTIVE

Amir

"3,462 to the nearest hundred is 3,400"

What went wrong?

Correct answer: _____

Bea

"0.0649 to the nearest hundredth is 0.07"

What went wrong?

Correct answer: _____

Caro

"My estimate was exactly right, so estimating is the same as working it out"

What went wrong?

Correct answer: _____

Task 1.6 • Think it through

CHALLENGE

a. Estimate $4,800 \div 19$ by rounding 19 to 20. Is your estimate too high or too low, and how do you know without calculating?

b. A whole number rounds to 400 to the nearest hundred. What is the smallest it could be? _____ The largest? _____

c. Why do shops price things at \$9.99 instead of \$10? _____

d. Estimate how many minutes you have been alive. Show the rounding you used. _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Close Enough

ANSWERS

Task 1.1 • Round it

- | | | | | | |
|----------|--------------------|--------------|-----------|---------------------|-------------|
| a. 68 | → nearest ten | 70 | e. 8.76 | → nearest tenth | 8.8 |
| b. 342 | → nearest hundred | 300 | f. 0.0847 | → nearest hundredth | 0.08 |
| c. 1,650 | → nearest hundred | 1,700 | g. 12.5 | → nearest whole | 13 |
| d. 4,499 | → nearest thousand | 4,000 | h. 249.5 | → nearest ten | 250 |

d is the trap. 4,499 looks close to 5,000, but rounding to the thousand means looking only at the hundreds digit, which is 4. The 99 is irrelevant. In f the hundredths digit is 6 and the next digit is 4, so it stays at 0.08.

Task 1.2 • Estimate first

- | | | |
|-------------------|--------------|---------|
| a. $40 + 50$ | = 90 | (89) |
| b. $400 - 200$ | = 200 | (225) |
| c. 20×20 | = 400 | (399) |
| d. 7×8 | = 56 | (55.89) |
| e. $600 \div 30$ | = 20 | (20.24) |
| f. 3×50 | = 150 | (153.6) |

The exact answers are in brackets for your reference only — students were told not to calculate them. Accept 3×48 or 3.2×50 in f; any sensible rounding is a correct method.

Task 1.3 • The trolley

- $\$5 + \$12 + \$3 + \$7 + \$19 + \3
- | | |
|------------------------|------------------|
| a. estimate | = \$49 |
| b. yes, \$50 covers it | |
| c. exact | = \$49.39 |
| d. out by | 39 cents |

e is the real question. Rounding every price up would give \$50 and a guaranteed safe answer — you can never be short. Rounding to the nearest is more accurate but can land you under. Which you choose depends on whether being wrong costs you anything, and at a till it does.

Task 1.4 • Is that even possible?

- | | | |
|---------------------------|-----------|-----------------------------------|
| a. $4.2 \times 19 = 79.8$ | ok | $\approx 4 \times 20 = 80$ |
| b. $312 \div 6 = 5.2$ | no | $\approx 300 \div 6 = 50$, so 52 |
| c. $0.5 \times 88 = 176$ | no | half of 88 is 44 |
| d. $9.8 + 10.4 = 20.2$ | ok | $\approx 10 + 10 = 20$ |
| e. $1,000 - 249 = 751$ | ok | $\approx 1,000 - 250 = 750$ |
| f. $24 \times 0.9 = 216$ | no | 0.9 of 24 is just under 24 |

b and f are both decimal-point slips of exactly one place, which is the single most common calculator error there is. c is the one worth talking about: multiplying by a number less than 1 always makes the answer *smaller*, and a student who feels that in their bones will never make the mistake again.

Task 1.4b has no single right answer, only better and worse reasoning. a: the nearest thousand — nobody counts a crowd to the person. b: leave it exactly, because a dose is not a place to be approximately right. c: 150 million km, and astronomers really do write it that way. d: about 19 cents, but a shop would quote 18.5c because it sells them in hundreds.

Task 1.5 • The three errors

Amir — 3,500. He looked at the wrong digit. Rounding to the hundred means checking the tens digit, which is 6.

Bea — 0.06. The hundredths digit is 6 and the digit to its right is 4, so it stays. The 9 further along changes nothing.

Caro — no. Her estimate matching was luck. An estimate is a check on an exact calculation, not a replacement for one — which is exactly what Task 1.4 was for.

Task 1.6 • Challenge

a. Too low, and you can say so without calculating. $4,800 \div 20 = 240$. Dividing by 20 instead of 19 means sharing between *more* parts, so each part comes out smaller. The real answer is about 253.

b. 350 and 449. Nearly everyone gets 350; the top end catches people, who say 450 — but 450 rounds up to 500.

c. Because a shopper reading left to right registers the 9 first and files it as “nine dollars something”. It is rounding used against you.

d. Roughly 7 million minutes for a 13-year-old: $13 \times 365 \times 24 \times 60$, or about $13 \times 400 \times 500 = 2.6$ million ... accept anything in the millions with visible rounding. The exact figure is about 6.8 million; the method matters far more than the number.

Point of Order

2 MINUTES

Students multiply and divide decimals, learn the count-the-places rule and why it works, then meet the two facts that overturn what everybody assumes: multiplying can make a number smaller, and dividing can make it bigger.

You do not need to know this subject

Multiply as though the decimal points were not there, then count places. Divide by shifting both numbers until the divisor is whole. Both rules are printed on the student page and every step is in the key.

Photocopy these pages only

Pages 12–14

One per student. Do not photocopy this page or the answer key.

What students need

A pencil. No calculators — a calculator hides exactly the step this lesson is about.

WHAT TO SAY TO START

“Eight times a half is four. You multiplied and the answer got smaller. That is not a trick and it is not a mistake, and by the end of today you will know exactly when it happens and why.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the two rules	Count the places
14–28	Task 2.1 — multiply and divide	Steady practice
28–40	Task 2.3 — money and measurement	Real prices
40–50	Task 2.4 — bigger or smaller?	The surprise of the lesson
50–55	Challenge, collect sheets	Discussion question below

Questions students will ask

“Where does the decimal point go?”

Count the digits after the point in both numbers you multiplied, add them, and the answer has that many. 0.3×0.4 has one and one, so the answer has two: 0.12.

“Why does $0.3 \times 0.4 = 0.12$ and not 1.2?”

Because you are taking three tenths of four tenths, and that has to be far less than either. Estimation catches it: a bit of a bit is a very small bit.

“How do I divide by a decimal?”

Shift the point in both numbers the same number of places until the one you are dividing by is a whole number. $7.5 \div 0.5$ becomes $75 \div 5$, which is 15.

If they finish early

Task 2.6 asks which is bigger, $12 \div 0.4$ or 12×0.4 . The gap is enormous and almost nobody predicts it correctly first time.

Curriculum focus • for the regular teacher’s records

Grade 6–7: fluently multiplying and dividing multi-digit decimals using the standard algorithm. Discussion question for the last five minutes: “How can multiplying two numbers give an answer smaller than both of them?”

Where the point goes

UNDERSTAND

16 MIN

NAME _____

CLASS _____

DATE _____

Two rules, one idea

MULTIPLYING – count places

Ignore the points: $3 \times 4 = 12$

0.3 has 1 place

0.4 has 1 place

$1 + 1 = 2$ places

→ $0.3 \times 0.4 = 0.12$

DIVIDING – shift both

$7.5 \div 0.5$

shift both 1 place right

$= 75 \div 5$

$= 15$

The answer does not change
because you scaled both
numbers the same way.

THE CHECK THAT SAVES YOU

6.9×8.1

$\approx 7 \times 8 = 56$

so **55.89**, not 5.589

and not 558.9

Estimate first. Every
time.

Task 2.1 • Multiply

DO

#	CALCULATION	ANSWER	#	CALCULATION	ANSWER
a	0.6×4		e	0.07×3	
b	0.3×0.4		f	1.5×1.5	
c	1.2×5		g	0.25×12	
d	2.5×0.8		h	6.4×0.5	

Task 2.2 • Divide

DO

Shift both numbers until you are dividing by a whole number.

#	CALCULATION	BECOMES	ANSWER	#	CALCULATION	BECOMES	ANSWER
a	$4.8 \div 2$			d	$1.2 \div 0.4$		
b	$0.36 \div 3$			e	$9 \div 0.25$		
c	$7.5 \div 0.5$			f	$0.144 \div 1.2$		

Money, and a surprise

KEY SKILL

20 MIN

Task 2.3 • Real prices, real measurements

APPLY

Estimate first in the margin, then work it out exactly.

a. Petrol is \$1.85 a litre. What does 42 litres cost?

b. Six concert tickets cost \$103.50 altogether. What is one ticket?

c. Cheese is \$18.40 a kilogram. What does 0.35 kg cost?

d. A 2.5 m board is cut into 0.4 m pieces. How many pieces, and what is left over?

Task 2.4 • Bigger or smaller?

PREDICT

Before you calculate anything, circle whether the answer will be bigger or smaller than the number in bold. Then work it out and see.

#	CALCULATION	BIGGER OR SMALLER?	ANSWER
a	8 × 0.5	bigger / smaller	
b	8 × 1.5	bigger / smaller	
c	8 ÷ 0.5	bigger / smaller	
d	8 ÷ 2	bigger / smaller	
e	0.6 × 0.6	bigger / smaller	

Finish the rule: multiplying by a number less than 1 always makes the answer _____, and dividing by a number less than 1 always makes it _____.

Errors and edges

STRETCH

17 MIN

Task 2.5 • Find the error

DETECTIVE

Dan

$$0.4 \times 0.5$$

$$= 2.0$$

What went wrong?

Correct answer: _____

Eve

$$3.6 \div 0.6$$

$$= 0.6$$

What went wrong?

Correct answer: _____

Finn

“Multiplying always
makes a number
bigger”

What went wrong?

Correct answer: _____

Task 2.6 • Think it through

CHALLENGE

- a. Which is bigger, $12 \div 0.4$ or 12×0.4 ? Work out both. _____
- b. $0.2 \times 0.2 = 0.04$, which is smaller than both. Explain in words why that has to happen. _____
- c. Fill the gap: _____ $\times 0.25 = 9$. _____
- d. A shop sells rope at \$2.40 a metre and you have \$18. How much rope can you buy? _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Point of Order

ANSWERS

Task 2.1 • Multiply

- a. $0.6 \times 4 = 2.4$
- b. $0.3 \times 0.4 = 0.12$
- c. $1.2 \times 5 = 6$
- d. $2.5 \times 0.8 = 2$
- e. $0.07 \times 3 = 0.21$
- f. $1.5 \times 1.5 = 2.25$
- g. $0.25 \times 12 = 3$
- h. $6.4 \times 0.5 = 3.2$

c, d and g come out whole, which students often think means they have gone wrong. They have not — 2.5×0.8 really is exactly 2. Accept 2.0 and 6.0.

Task 2.2 • Divide

- a. $4.8 \div 2 = 2.4$
- b. $0.36 \div 3 = 0.12$
- c. $75 \div 5 = 15$
- d. $12 \div 4 = 3$
- e. $900 \div 25 = 36$
- f. $1.44 \div 12 = 0.12$

c, d, e and f all needed the shift. In e, $9 \div 0.25$ means “how many quarters are in 9?” — four in every whole, so thirty-six. That reading of division is worth saying aloud.

Task 2.3 • Real prices

- a. $1.85 \times 42 = \$77.70$ (estimate $\approx 2 \times 40 = \$80$)
- b. $103.50 \div 6 = \$17.25$ (estimate $\approx 100 \div 6 \approx \17)
- c. $18.40 \times 0.35 = \$6.44$ (estimate $\approx 18 \times \frac{1}{3} = \6)
- d. $2.5 \div 0.4 = 6$ pieces, with 0.1 m left over

d is the one that separates arithmetic from thinking. $2.5 \div 0.4 = 6.25$, but you cannot cut a quarter of a piece and call it a piece. Six whole pieces use 2.4 m, leaving 10 cm. A student who writes 6.25 pieces has done the division and not the question.

Task 2.4 • Bigger or smaller?

- a. 8×0.5 **smaller** = 4
- b. 8×1.5 **bigger** = 12
- c. $8 \div 0.5$ **bigger** = 16
- d. $8 \div 2$ **smaller** = 4
- e. 0.6×0.6 **smaller** = 0.36

The rule they should finish with: multiplying by less than 1 makes it **smaller**, dividing by less than 1 makes it **bigger**. Compare a and c — same two numbers, and the answers are 4 and 16.

Task 2.5 • The three errors

Dan — 0.20. He got $4 \times 5 = 20$ right and then placed the point by guesswork. Two decimal places in, two out.

Eve — 6. Shifting both gives $36 \div 6$. Her answer is smaller than the number she started with, which dividing by 0.6 can never do.

Finn — no. $8 \times 0.5 = 4$. Multiplying only makes a number bigger if you multiply by more than 1, which is true for every whole number he has ever used, which is exactly why the belief forms.

Task 2.6 • Challenge

- a. $12 \div 0.4 = 30$ and $12 \times 0.4 = 4.8$ — division wins, by a distance
- c. 36 d. $18 \div 2.40 = 7.5$ metres

b. Taking 0.2 of something means taking a fifth of it. Taking a fifth of a fifth leaves you with a twenty-fifth, which is far less than either. Any answer along the lines of “a part of a part is a smaller part” is complete.

d is worth a mention: 7.5 m is a perfectly sensible answer here because rope can be cut anywhere — unlike the six boards in Task 2.3.

Change of Units

2 MINUTES

Students convert within the metric system, convert common customary units, compare quantities given in different units, and solve real problems where the units do not match. It closes on the one conversion almost everybody gets wrong: square metres to square centimetres.

You do not need to know this subject

Metric conversions are all multiplying or dividing by 10, 100 or 1,000, and the ladder is printed on the student page. The customary conversions students need are printed there too.

Photocopy these pages only

Pages 17–19

One per student. Do not photocopy this page or the answer key.

What students need

A pencil. A calculator is fine if you have one; none of the arithmetic needs it.

WHAT TO SAY TO START

“A recipe asks for 1.5 kilograms of flour and you have three 400 gram bags. Are you short? You cannot answer that until both numbers are in the same units, and that one step is most of what today is about.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the ladder panel	Bigger unit to smaller means multiply
14–26	Task 3.1 — metric and customary	Steady practice
26–36	Task 3.2 — which is bigger?	Convert, then compare
36–48	Task 3.3 — real problems	Match the units first
48–55	Challenge, collect sheets	Discussion question below

Questions students will ask

“Do I multiply or divide?”

Going to a smaller unit means you need more of them, so multiply. Going to a bigger unit means fewer, so divide. Metres to centimetres: more, so $\times 100$.

“How do I remember the metric prefixes?”

kilo is a thousand, centi is a hundredth, milli is a thousandth. They never change, whatever the unit is attached to.

“Why isn’t 1 square metre equal to 100 square centimetres?”

Because a square metre is 100 cm by 100 cm, which is $10,000 \text{ cm}^2$. Task 3.6 walks students through it and it surprises nearly all of them.

If they finish early

Task 3.6 asks how many litres are in a swimming pool 10 m by 5 m by 1.5 m. The answer is 75,000, and getting there needs both a volume and a unit conversion.

Curriculum focus • for the regular teacher’s records

Grade 6–7: converting measurement units within and between systems, and using unit reasoning to solve multi-step problems. Discussion question for the last five minutes: *“Why does the whole world except a few countries use the metric system?”*

Up and down the ladder

UNDERSTAND

15 MIN

NAME _____

CLASS _____

DATE _____

The ladder, and the two directions on it

METRIC

km $\times 1000$ m
 m $\times 100$ cm
 cm $\times 10$ mm
 kg $\times 1000$ g
 L $\times 1000$ mL
 t $\times 1000$ kg

← divide multiply →

CUSTOMARY

1 ft = 12 in
 1 yd = 3 ft
 1 lb = 16 oz
 1 gal = 4 qt
 1 h = 60 min
 1 min = 60 s

WHICH WAY?

Bigger unit → smaller unit
you need **more** of them
so **multiply**

3 km → m
 $3 \times 1000 = 3,000$ m

250 cm → m
 $250 \div 100 = 2.5$ m

Task 3.1 • Metric

DO

#	CONVERT	ANSWER	#	CONVERT	ANSWER
a	3 km → m		d	750 mL → L	
b	250 cm → m		e	45 mm → cm	
c	1.4 kg → g		f	0.8 t → kg	

Task 3.1b • Customary and time

DO

#	CONVERT	ANSWER	#	CONVERT	ANSWER
a	5 ft → in		d	36 in → yd	
b	3 lb → oz		e	2.5 h → min	
c	2 gal → qt		f	90 min → h	

Compare first, then convert

KEY SKILL

22 MIN

Task 3.2 • Which is bigger?

COMPARE

Convert one of them so both are in the same unit, then circle the larger.

#	FIRST	SECOND	WORKING	LARGER
a	1.2 kg	1,150 g		
b	2.5 L	2,400 mL		
c	45 cm	0.5 m		
d	500 g	1 lb		

For d you need to know that 1 lb is about 454 g.

Task 3.3 • When the units do not match

APPLY

Convert first. Then answer in a sentence, with the unit.

a. A recipe needs 1.5 kg of flour. You have three 400 g bags. Is that enough?

c. A room is 4.2 m by 3.5 m. Carpet costs \$46 per square metre. What is the total?

b. A tank holds 25 L. A jug holds 750 mL. How many full jugs fill it?

d. A trip takes 3 h 45 min at an average of 60 km/h. How far is it?

Task 3.4 • Read the label

APPLY

A bottle of cordial holds **1.25 L**. The label says one serve is **50 mL** of cordial topped up with water.

a. How many serves in a bottle? _____

c. A party needs 60 serves. How many bottles? _____

b. The bottle costs \$4.50. Cost per serve? _____

d. How much cordial is left over? _____

Errors and one big surprise

STRETCH

17 MIN

Task 3.5 • Find the error

DETECTIVE

Gina

“2.5 m = 25 cm”

What went wrong?

Correct answer: _____

Huy

“1,500 g = 15 kg”

What went wrong?

Correct answer: _____

Ivy“2 h 30 min
= 2.30 hours”

What went wrong?

Correct answer: _____

Task 3.6 • Think it through

CHALLENGE

- a. Which is heavier, 1 kg of feathers or 1,000 g of lead? _____
- b. A pool is 10 m long, 5 m wide and 1.5 m deep. How many litres does it hold? Remember $1 \text{ m}^3 = 1,000 \text{ L}$.

- c. A square metre is a square 1 m by 1 m. Draw it, mark each side in centimetres, and work out how many square centimetres are inside. It is not 100.

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Change of Units

ANSWERS

Task 3.1 • Metric

- a. $3 \times 1000 = 3,000$ m
- b. $250 \div 100 = 2.5$ m
- c. $1.4 \times 1000 = 1,400$ g
- d. $750 \div 1000 = 0.75$ L
- e. $45 \div 10 = 4.5$ cm
- f. $0.8 \times 1000 = 800$ kg

Task 3.1b • Customary and time

- a. $5 \times 12 = 60$ in
- b. $3 \times 16 = 48$ oz
- c. $2 \times 4 = 8$ qt
- d. $36 \div 36 = 1$ yd
- e. $2.5 \times 60 = 150$ min
- f. $90 \div 60 = 1.5$ h

d takes two steps — 36 in is 3 ft, and 3 ft is 1 yd. Accept either route.

Task 3.2 • Which is bigger?

- a. 1.2 kg = 1,200 g → **1.2 kg** is bigger than 1,150 g
- b. 2.5 L = 2,500 mL → **2.5 L** is bigger than 2,400 mL
- c. 0.5 m = 50 cm → **0.5 m** is bigger than 45 cm
- d. 1 lb ≈ 454 g → **500 g** is bigger than 1 lb

In every one of these the number that *looks* bigger is not, because it is written in a smaller unit. That is the whole point of the task, and it is worth saying out loud once the class has finished it.

Task 3.3 • Mismatched units

- a. $3 \times 400 = 1,200$ g vs 1,500 g
no — 300 g short
- b. 25 L = 25,000 mL
 $25,000 \div 750 = 33.3$
33 full jugs (250 mL left)
- c. $4.2 \times 3.5 = 14.7$ m²
 $14.7 \times 46 = \$676.20$
- d. 3 h 45 min = 3.75 h
 $3.75 \times 60 = 225$ km

d is the one to watch: 3 h 45 min is 3.75 hours, not 3.45. Students who write 3.45 get 207 km and it looks plausible, which is what makes the mistake dangerous.

Task 3.4 • Read the label

- a. 1.25 L = 1,250 mL
 $1,250 \div 50 = 25$ serves
- b. $4.50 \div 25 = 18$ cents a serve
- c. $60 \div 25 = 2.4$ → **3 bottles**
- d. 3 bottles = 75 serves
 $75 - 60 = 15$ serves
= 750 mL left over

c is a rounding-up question: 2.4 bottles is not a thing you can buy, and two bottles would leave the party ten serves short.

Task 3.5 • The three errors

Gina — 250 cm. She moved the point one place instead of two. Metres to centimetres is $\times 100$.

Huy — 1.5 kg. He divided by 100 instead of 1,000.

Ivy — 2.5 hours. Thirty minutes is half an hour, and half of 1 is 0.5, not 0.30. Time is the one measurement that is not decimal, which is exactly why this catches people.

Task 3.6 • Challenge

a. They weigh the same. 1 kg is 1,000 g. The question is a very old joke and it works because “feathers” and “lead” shout louder than the numbers.

b. 75,000 litres. $10 \times 5 \times 1.5 = 75$ m³, and each cubic metre is 1,000 L.

c. 10,000 cm². Each side is 100 cm, so the square holds 100×100 little squares. Almost every student first says 100. Ask them to draw a 10 cm $\times$ 10 cm square and count — 100 square centimetres, from sides of only 10. The jump to 10,000 then makes sense.

Both Directions

2 MINUTES

Students find opposites, evaluate absolute value, order negative numbers, and then apply the difference between “which is further from zero” and “which is larger” to temperatures, submarines, bank balances and elevations.

You do not need to know this subject

Absolute value just means distance from zero, and distance is never negative. That single sentence answers almost every question on the sheet, and the key shows every step.

Photocopy these pages only

Pages 22–24

One per student. Do not photocopy this page or the answer key.

What students need

A pencil. No calculators needed.

WHAT TO SAY TO START

“A submarine is 120 metres below the surface and a bird is 45 metres above it. Neither number is bigger than the other in the way you might expect, and working out how far apart they are needs an idea you will meet today.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–12	Read the number line panel	Distance from zero
12–24	Task 4.1 — opposites and absolute value	Fast and confidence-building
24–34	Task 4.2 — ordering and comparing	The pairs are deliberate
34–44	Task 4.3 — real contexts	Draw the number line
44–50	Challenge, collect sheets	Discussion question below

Questions students will ask

“What do the two lines mean?”

$|-7|$ is read “the absolute value of negative seven” and it means how far that number is from zero. Seven steps. So the answer is 7.

“Can an absolute value be negative?”

No, ever. It is a distance, and you cannot be minus five metres away from something. Task 4.5 asks students to explain exactly this.

“Is -7 bigger than -3 ?”

No. On a number line -7 is further left, so it is smaller. It is *further from zero*, which is a different question — and telling those two apart is the whole lesson.

If they finish early

Task 4.5 asks what x could be if $|x| = 6$. There are two answers, and students who find only one have found the easy half.

Curriculum focus • for the regular teacher’s records

Grade 6–7: understanding absolute value as distance from zero, ordering rational numbers, and interpreting statements of order in real-world contexts. Discussion question for the last five minutes: “When does “bigger” not mean “further from zero”?”

Distance, not direction

UNDERSTAND

14 MIN

NAME _____

CLASS _____

DATE _____

Two different questions about the same number

OPPOSITE

same distance, other side

opposite of 7 = -7

opposite of -12 = 12

opposite of 0 = 0

ABSOLUTE VALUE

distance from zero

$|8| = 8$

$|-8| = 8$

$|0| = 0$

never negative

 -7 is smaller than -3 — it is further left. $|-7|$ is bigger than $|-3|$ — it is further from zero.

Task 4.1 • Opposites

DO

#	NUMBER	ITS OPPOSITE	#	NUMBER	ITS OPPOSITE
a	7		d	3.5	
b	-12		e	$-\frac{2}{3}$	
c	0		f	$-(-5)$	

Task 4.2 • Absolute value

DO

#	EXPRESSION	VALUE	#	EXPRESSION	VALUE
a	$ 8 $		e	$ 15 - 20 $	
b	$ -8 $		f	$ -7 + -2 $	
c	$ 0 $		g	$ -7 + 2 $	
d	$ -3.5 $		h	$- -6 $	

Look at **f** and **g**. Why are the answers different when the numbers are the same? _____

Ordering, depths and balances

KEY SKILL

20 MIN

Task 4.3 • Order and compare

DO

a. Write these in order, smallest first: -5 3 -1 0 4 -3

b. Put $<$ or $>$ in each gap.

#	COMPARE		< OR >
i	-5	-2	
ii	$ -5 $	$ -2 $	
iii	-8	3	

#	COMPARE		< OR >
iv	$ -8 $	$ 3 $	
v	-1	-1.5	
vi	$ -1 $	$ -1.5 $	

Task 4.4 • Real contexts

APPLY

Sketch a number line for each one before you answer.

a. The temperature drops from 4°C to -6°C . By how many degrees?

c. A bank balance is $-\$85$ and you deposit $\$200$. What is the new balance?

b. A submarine sits at -120 m and a bird flies at 45 m. How far apart?

d. Town A is at 300 m elevation, Town B at -20 m. What is the difference?

Comparing and concluding

STRETCH

20 MIN

Task 4.5 • Which is further from zero?

COMPARE

For each pair, say which number is **larger**, and separately which is **further from zero**. Sometimes they are the same number and sometimes they are not.

#	PAIR	LARGER	FURTHER FROM ZERO
a	-9 and 4		
b	6 and 2		
c	-3 and -7		
d	0 and -5		

In which rows did the two answers disagree, and what did those rows have in common? _____

Task 4.6 • Find the error

DETECTIVE

Jai

$$|-9| = -9$$

What went wrong?

Correct answer: _____

Kim

"-7 is bigger than
-3 because
 $7 > 3$ "

What went wrong?

Correct answer: _____

Lou

"The opposite of
-5 is $-(-5)$,
which is -5"

What went wrong?

Correct answer: _____

Task 4.7 • Think it through

CHALLENGE

- If $|x| = 6$, what could x be? _____
- Can $|x|$ ever be negative? Explain. _____
- If $|x| = x$, what do you know about x ? _____
- Find a value of x for which $-x$ is positive. _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Both Directions

ANSWERS

Task 4.1 • Opposites

- a. 7 → -7 d. 3.5 → -3.5
b. -12 → 12 e. $-\frac{2}{3}$ → $\frac{2}{3}$
c. 0 → 0 f. -(-5) → -5

f needs unpacking: $-(-5)$ is already 5, so its opposite is -5. c is the only number that is its own opposite.

Task 4.2 • Absolute value

- a. $|8|$ = 8
b. $|-8|$ = 8
c. $|0|$ = 0
d. $|-3.5|$ = 3.5
e. $|15-20|$ = $|-5|$ = 5
f. $7 + 2$ = 9
g. $|-5|$ = 5
h. $-(6)$ = -6

f and g are the point. In f each number is measured from zero first and then added. In g they are added first and the result measured. h is the one case where a minus sign survives, because it sits outside the bars.

Task 4.3 • Order and compare

- a. -5, -3, -1, 0, 3, 4

- i. $-5 < -2$ iv. $|-8| > |3|$
ii. $|-5| > |-2|$ v. $-1 > -1.5$
iii. $-8 < 3$ vi. $|-1| < |-1.5|$

Every pair is deliberately the same two numbers with and without the bars, and in every single case the symbol flips. That is the whole lesson in six lines: with negatives, being further from zero and being larger are opposite things.

Task 4.4 • Real contexts

- a. 4 to -6 → 10 degrees
b. $45 - (-120)$ → 165 m apart
c. $-85 + 200$ → \$115
d. $300 - (-20)$ → 320 m

Every one of these is really $|\text{first} - \text{second}|$: the distance between two points on a line. Students who count the steps on a sketch get them all right; students who try to remember a rule about two minus signs do not.

Task 4.5 • Larger vs further from zero

- a. -9 and 4 larger 4 further -9
b. 6 and 2 larger 6 further 6
c. -3 and -7 larger -3 further -7
d. 0 and -5 larger 0 further -5

a, c and d disagree; b does not. What the disagreeing rows share is a negative number. Among positives the two questions always give the same answer, which is why the distinction never comes up until negatives arrive.

Task 4.6 • The three errors

Jai — 9. Absolute value is a distance and distances are never negative.

Kim — no, -7 is smaller. She is comparing distances from zero, not positions. -7 is further left on the line, so it is less.

Lou — 5. Two minus signs cancel: $-(-5) = 5$.

Task 4.7 • Challenge

a. $x = 6$ or $x = -6$. Both are six steps from zero. A student who gives only 6 has done half the question.

b. No. It measures distance from zero, and nothing can be a negative distance away from anything.

c. x is zero or positive. A number equals its own distance from zero only when it is not negative.

d. Any negative x . If $x = -4$ then $-x = 4$. This is worth dwelling on: the minus sign in front of a letter does not mean the value is negative, and that misreading causes real trouble in algebra later.

Balancing Act

2 MINUTES

Students solve equations with the unknown on both sides, expand brackets first where needed, and then meet the two strange cases: an equation every number satisfies and an equation no number satisfies. It ends with two gym memberships and the month at which the cheaper one changes.

You do not need to know this subject

Every answer can be checked by substituting it back in. If both sides come out the same number, the student is right, whatever route they took. The key shows every line of working.

Photocopy these pages only

Pages 27–29

One per student. Do not photocopy this page or the answer key.

What students need

A pencil. No calculators needed.

WHAT TO SAY TO START

“Everything you have solved so far had the letter on one side and the numbers on the other. Today there is a letter on both sides, and the whole game is getting them together — which turns out to be one extra move, not a new subject.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the balance panel	Whatever you do, do to both sides
14–28	Task 5.1 — straight solving	Move the smaller x term
28–38	Task 5.2 — brackets first	Expand, then collect
38–48	Task 5.3 and 5.4 — strange cases and gyms	Let them be surprised
48–55	Challenge, collect sheets	Discussion question below

Questions students will ask

“Which side should I move the x to?”

It does not matter mathematically, but moving the *smaller* x term keeps the coefficient positive and avoids a negative later. That tip is on the student page.

“How do I know my answer is right?”

Put it back into the original equation. Both sides must give the same number. This is the single most useful habit in the lesson and it needs no teacher.

“What if the x disappears completely?”

Then look at what is left. If it is true, like $8 = 8$, every number works. If it is false, like $3 = 7$, no number does. Task 5.3 is entirely about this.

If they finish early

Task 5.6 asks students to invent an equation with x on both sides whose answer is exactly 5. There are infinitely many and building one backwards is harder than solving one.

Curriculum focus • for the regular teacher’s records

Grade 7–8: solving linear equations with rational coefficients, including equations whose solutions require expanding brackets and collecting like terms. Discussion question for the last five minutes: *“How can an equation have no answer at all?”*

Getting the letters together

UNDERSTAND

18 MIN

NAME _____

CLASS _____

DATE _____

One extra move, then it is an ordinary equation

$$5x = 3x + 8$$

Take $3x$ from **both** sides:

$$2x = 8$$

$$x = 4$$

Check: $5(4) = 20$
 $3(4) + 8 = 20 \checkmark$

$$7x - 2 = 5x + 6$$

 $-5x$ both sides:

$$2x - 2 = 6$$

 $+2$ both sides:

$$2x = 8$$

$$x = 4$$

TIPMove the **smaller** x term.

In $2x + 9 = 6x - 3$, take $2x$ from both sides, not $6x$.
 You get $4x$, not $-4x$, and you avoid a sign error.

Task 5.1 • Solve, then check

DO

Substitute your answer back into the original equation and tick it if both sides match.

#	EQUATION	WORKING	X =	#	EQUATION	WORKING	X =
a	$5x = 3x + 8$			d	$2x + 9 = 6x - 3$		
b	$7x - 2 = 5x + 6$			e	$8 - x = 2x + 2$		
c	$4x + 1 = x + 10$			f	$3x - 5 = 7x + 11$		

Task 5.2 • Brackets first

DO

Expand every bracket before you move anything.

#	EQUATION	EXPANDED	X =	#	EQUATION	EXPANDED	X =
a	$4(x + 2) = 3x + 11$			c	$2(2x + 3) = 5x - 1$		
b	$3(x - 4) = 2(x + 1)$			d	$5(x - 1) = 3(x + 3)$		

Strange cases and real choices

KEY SKILL

20 MIN

Task 5.3 • When the x vanishes

THINK

Solve each one. In two of them every x term disappears. When that happens, look at what is left and decide whether it is true or false.

#	EQUATION	WHAT IS LEFT	ONE ANSWER / EVERY NUMBER / NONE
a	$x + 8 = 8 + x$		
b	$2x + 3 = 2x + 7$		
c	$3(x + 2) = 3x + 6$		
d	$4x - 1 = 4x + 1$		
e	$6x + 2 = 4x + 10$		

Task 5.4 • Two gyms

APPLY

Gym A charges a \$40 joining fee and \$15 a month. Gym B charges \$10 to join and \$20 a month.

MONTHS	GYM A	GYM B
1		
3		
6		
12		

a. Write a rule for each gym. _____

b. Set them equal and solve. After how many months do they cost the same? _____

c. What is the cost at that point? _____

d. Which gym is better for someone joining for a whole year?

Cards, candles and invention

STRETCH

17 MIN

Task 5.5 • Two more situations

APPLY

a. Ana has three times as many cards as Ben. If Ana gives Ben 12 cards they have the same number. How many does each have to begin with?

b. One candle is 20 cm tall and burns 2 cm an hour. Another is 15 cm tall and burns 1 cm an hour. When are they the same height, and how tall are they then?

Task 5.6 • Find the error

DETECTIVE

Mia

$$5x = 3x + 8$$

$$8x = 8$$

$$x = 1$$

What went wrong?

Correct answer: _____

Nate

$$2(x + 3) = x + 10$$

$$2x + 3 = x + 10$$

$$x = 7$$

What went wrong?

Correct answer: _____

Omar

$$2x + 3 = 2x + 7$$

$$3 = 7$$

$$\text{so } x = 0$$

What went wrong?

Correct answer: _____

Task 5.7 • Think it through

CHALLENGE

a. Invent an equation with x on both sides whose answer is $x = 5$. _____

b. In $4x + a = 2x + 10$ the answer is $x = 3$. Find a . _____

c. Why does $x + 8 = 8 + x$ work for every number? _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Balancing Act

ANSWERS

Task 5.1 • Solve and check

- | | |
|---------------|----------|
| a. $2x = 8$ | $x = 4$ |
| b. $2x = 8$ | $x = 4$ |
| c. $3x = 9$ | $x = 3$ |
| d. $12 = 4x$ | $x = 3$ |
| e. $6 = 3x$ | $x = 2$ |
| f. $-16 = 4x$ | $x = -4$ |

In d and e the x ended up on the right, which is fine and is what the tip in the panel produces. f is the only negative answer; check it by substituting: $3(-4) - 5 = -17$ and $7(-4) + 11 = -17$. ✓

Task 5.2 • Brackets first

- | | |
|-----------------------|----------|
| a. $4x + 8 = 3x + 11$ | $x = 3$ |
| b. $3x - 12 = 2x + 2$ | $x = 14$ |
| c. $4x + 6 = 5x - 1$ | $x = 7$ |
| d. $5x - 5 = 3x + 9$ | $x = 7$ |

b gives a surprisingly large answer, which makes students doubt themselves. Check it: $3(10) = 30$ and $2(15) = 30$. ✓

Task 5.3 • When the x vanishes

- | | | |
|----------------------|----------------------|--------------------|
| a. $x + 8 = 8 + x$ | $\rightarrow 8 = 8$ | every number works |
| b. $2x + 3 = 2x + 7$ | $\rightarrow 3 = 7$ | no solution |
| c. $3x + 6 = 3x + 6$ | $\rightarrow 6 = 6$ | every number works |
| d. $4x - 1 = 4x + 1$ | $\rightarrow -1 = 1$ | no solution |
| e. $2x = 8$ | $\rightarrow x = 4$ | one answer |

a and c are the same statement written two ways — both sides are genuinely identical, so any number balances them. b and d ask for a number that is both three and seven more than the same thing, which nothing can be. Students find “no solution” unsatisfying; it is worth saying that finding out something is impossible is a real result, not a failure to finish.

Task 5.4 • Two gyms

months	1	3	6	12
Gym A	\$55	\$85	\$130	\$220
Gym B	\$30	\$70	\$130	\$250

- a. A: $40 + 15m$ B: $10 + 20m$
 b. $40 + 15m = 10 + 20m$
 $30 = 5m \rightarrow 6 \text{ months}$
 c. **\$130** at either gym
 d. **Gym A** — \$220 against \$250

The joining fee is the starting value and the monthly charge is the rate. Under six months B wins; over six months A does. Nobody is being ripped off — the two gyms are simply priced for different customers.

Task 5.5 • Cards and candles

- a. Ana = $3B$
 $3B - 12 = B + 12$
 $2B = 24 \rightarrow B = 12$
Ana 36, Ben 12
 check: 24 and 24 ✓
- b. $20 - 2t = 15 - t$
 $5 = t \rightarrow \text{after 5 hours}$
 both are **10 cm**

In b the taller candle is losing height faster, which is why they meet at all. Ask what happens after 10 hours — the first candle is gone and the equation stops describing anything real.

Task 5.6 • The three errors

Mia — $x = 4$. She added $3x$ to both sides instead of subtracting it. Her answer fails the check: $5(1) = 5$ but $3(1) + 8 = 11$.

Nate — $x = 4$. He multiplied the x by 2 and forgot the 3. The line should read $2x + 6 = x + 10$.

Omar — **there is no solution**. Reaching $3 = 7$ does not mean x is zero; it means no value of x can make the equation true. Substituting 0 gives $3 = 7$, which is still false.

Task 5.7 • Challenge

- a. Anything that checks out — $4x = 2x + 10$, or $7x - 3 = 5x + 7$. The reliable method is to start from $x = 5$, pick a left side such as $4x$, work out its value (20), and build a right side that also gives 20 at $x = 5$.
- b. **a = 4**. At $x = 3$ the left is $12 + a$ and the right is 16, so $a = 4$.
- c. Because the two sides are the same expression written in a different order. Adding 8 to a number and adding a number to 8 are the same operation, so the equation is true before you know anything about x at all.

Round Numbers

2 MINUTES

Students calculate circumference and area from a radius or a diameter, work backwards from a circumference to a radius, and apply both to a bicycle wheel, a pizza and a running track. It finishes by asking which pizza is actually better value, and the answer is not the obvious one.

You do not need to know this subject

Two formulas, both printed on the student page, and π is given as 3.14 throughout. Every calculation in the key is shown in full.

Photocopy these pages only

Pages 32–34

One per student. Do not photocopy this page or the answer key.

What students need

A pencil. A calculator is genuinely useful here — allow one if you have it.

WHAT TO SAY TO START

“Your bicycle wheel is about 66 centimetres across. Every time it turns, you travel the distance around its edge. Today you work out how far that is, and how many turns it takes to ride a kilometre.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the two formulas	Radius, not diameter, in both
14–26	Task 6.1 — straight calculation	Halve the diameter first
26–36	Task 6.2 — working backwards	Undo the formula
36–48	Task 6.3 — wheels, pizzas and tracks	Real numbers
48–55	Challenge, collect sheets	Discussion question below

Questions students will ask

“Do I use the radius or the diameter?”

The radius, in both formulas. If a question gives you the diameter, halve it first. That single step accounts for most of the wrong answers on this sheet.

“What is π ?”

The number of diameters that fit around the edge of any circle, and it is the same for every circle that has ever existed. It is about 3.14. Use 3.14 throughout.

“Why is it r^2 and not $2r$?”

Because area covers a surface, so it needs two lengths multiplied. Task 6.5 has a student who makes exactly this substitution and gets a third of the right answer.

If they finish early

Task 6.6 asks whether one 30 cm pizza at \$18 beats two 20 cm pizzas at \$22. The single large one wins comfortably and almost nobody predicts it.

Curriculum focus • for the regular teacher’s records

Grade 7: knowing and using the formulas for the area and circumference of a circle, and solving problems involving them. Discussion question for the last five minutes: *“Why does doubling the radius do something different to the circumference than to the area?”*

Round the edge, inside the edge

UNDERSTAND

18 MIN

NAME _____

CLASS _____

DATE _____

Two formulas, one radius

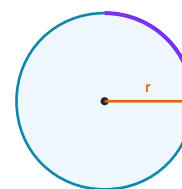
 $C = 2\pi r$ the distance round $A = \pi r^2$ the space insideuse $\pi \approx 3.14$ If you are given the
diameter, halve it first. $d = 20 \rightarrow r = 10$ $r = 5 \text{ cm}$

$$C = 2 \times 3.14 \times 5$$

$$= 31.4 \text{ cm}$$

$$A = 3.14 \times 5 \times 5$$

$$= 78.5 \text{ cm}^2$$

length $\rightarrow \text{cm}$
area $\rightarrow \text{cm}^2$ 

Task 6.1 • Circumference and area

DO

Two decimal places is plenty. Watch the units.

#	GIVEN	RADIUS	$C = 2\pi r$	$A = \pi r^2$
a	$r = 3 \text{ cm}$	3 cm		
b	$r = 5 \text{ cm}$			
c	$r = 10 \text{ m}$			
d	$d = 8 \text{ cm}$			
e	$d = 14 \text{ cm}$			
f	$d = 12 \text{ m}$			

Task 6.2 • Work backwards

REVERSE

Here you are given the answer and asked for the radius. Undo the formula one step at a time.

#	GIVEN	RADIUS	#	GIVEN	RADIUS
a	$C = 31.4 \text{ cm}$		c	$A = 78.5 \text{ cm}^2$	
b	$C = 18.84 \text{ cm}$		d	$A = 12.56 \text{ cm}^2$	

Real circles

20 MIN

Task 6.3 • Wheels, pizzas and tracks

APPLY

a. A bicycle wheel is 66 cm across. How far does the bike travel in one full turn?

c. A pizza is 30 cm across. What is the area of the top?

b. How many turns does that wheel make over one kilometre?

d. A running track has an inner circle of radius 30 m. The next lane out is radius 32 m. How much further is one lap in the outer lane?

Task 6.4 • A shape with a curved end

APPLY

A swimming lane marker is a rectangle **10 m by 6 m** with a **semicircle** of diameter 6 m joined to one short end.

a. Area of the rectangle? _____

b. Area of the semicircle? _____

c. Total area? _____

d. Length of the curved edge? _____

e. Perimeter of the whole shape? _____

Careful in e — one side of the rectangle is no longer on the outside.

Pizza economics

STRETCH

17 MIN

Task 6.5 • Find the error

DETECTIVE

Pia

$$\begin{aligned} r &= 6 \text{ cm} \\ A &= 3.14 \times 12 \\ &= 37.68 \end{aligned}$$

What went wrong?

Correct answer: _____

Quinn

$$\begin{aligned} d &= 10 \text{ cm} \\ A &= 3.14 \times 10^2 \\ &= 314 \end{aligned}$$

What went wrong?

Correct answer: _____

Raj

$$C = 2\pi r^2$$

What went wrong?

Correct answer: _____

Task 6.6 • Think it through

CHALLENGE

a. One pizza 30 cm across costs \$18. Two pizzas 20 cm across cost \$22 for the pair. Which gives you more pizza for your money? Show the working.

b. If you double the radius, what happens to the circumference? And to the area? _____

c. A circle's circumference and area come to the same number. What is its radius? _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Round Numbers

ANSWERS

Task 6.1 • Circumference and area

- a. $r = 3$ $C = 18.84 \text{ cm}$ $A = 28.26 \text{ cm}^2$
 b. $r = 5$ $C = 31.4 \text{ cm}$ $A = 78.5 \text{ cm}^2$
 c. $r = 10$ $C = 62.8 \text{ m}$ $A = 314 \text{ m}^2$
 d. $d = 8 \rightarrow r = 4$ $C = 25.12 \text{ cm}$ $A = 50.24 \text{ cm}^2$
 e. $d = 14 \rightarrow r = 7$ $C = 43.96 \text{ cm}$ $A = 153.86 \text{ cm}^2$
 f. $d = 12 \rightarrow r = 6$ $C = 37.68 \text{ m}$ $A = 113.04 \text{ m}^2$

Rows d, e and f all needed halving first. A student who forgets gets exactly four times the correct area, which is a distinctive enough error that you can diagnose it at a glance.

Task 6.2 • Work backwards

- a. $31.4 \div 3.14 = 10, \div 2 \rightarrow r = 5 \text{ cm}$
 b. $18.84 \div 6.28 \rightarrow r = 3 \text{ cm}$
 c. $78.5 \div 3.14 = 25, \sqrt{} \rightarrow r = 5 \text{ cm}$
 d. $12.56 \div 3.14 = 4, \sqrt{} \rightarrow r = 2 \text{ cm}$

c and d need a square root at the end, because the area formula squared the radius on the way in. Undoing happens in reverse order.

Task 6.5 • The three errors

Pia — 113.04 cm^2 . She doubled the radius instead of squaring it. r^2 means 6×6 , not 6×2 .

Quinn — 78.5 cm^2 . He used the diameter as the radius. Halve it first: $r = 5$.

Raj — $C = 2\pi r$. No square. The squared version gives an area, and an area cannot be a distance.

Task 6.3 • Wheels, pizzas, tracks

- a. $C = 3.14 \times 66$
 $= 207.24 \text{ cm}$ ($\approx 2.07 \text{ m}$)
 b. $100,000 \div 207.24$
 $\approx 483 \text{ turns}$
 c. $r = 15, A = 3.14 \times 225$
 $= 706.5 \text{ cm}^2$
 d. $2 \times 3.14 \times (32 - 30)$
 $= 12.56 \text{ m further}$

d is the elegant one. The extra distance depends only on the *difference* in radius, not on how big the track is — which is exactly why staggered starts on an athletics track are always the same offset whatever the stadium.

Task 6.4 • The curved end

- a. $10 \times 6 = 60 \text{ m}^2$
 b. $r = 3, \frac{1}{2} \times 3.14 \times 9 = 14.13 \text{ m}^2$
 c. $60 + 14.13 = 74.13 \text{ m}^2$
 d. $\frac{1}{2} \times 3.14 \times 6 = 9.42 \text{ m}$
 e. $10 + 6 + 10 + 9.42 = 35.42 \text{ m}$

In e the short end where the semicircle joins is inside the shape and does not count. Students who add all four rectangle sides get 41.42 and should be sent back to trace the outline with a finger.

Task 6.6 • Challenge

- a. one 30 cm: $r = 15, A = 706.5 \text{ cm}^2$ for \$18 $\rightarrow 39.3 \text{ cm}^2$ per dollar
 two 20 cm: $r = 10, A = 314$ each = 628 cm^2 for \$22 $\rightarrow 28.5 \text{ cm}^2$ per dollar
 $\rightarrow$ the single large pizza wins, and not narrowly

a surprises everyone, because two pizzas *sounds* like more than one. Two 20 cm pizzas have less pizza in them than one 30 cm, even though $20 + 20$ is more than 30. Adding diameters tells you nothing; area is what you eat.

b. **The circumference doubles and the area quadruples.** C has one r in it and A has two, so doubling r doubles C once and doubles A twice.

c. $r = 2. 2\pi r = \pi r^2$ gives $2 = r$. Note this is a coincidence of the numbers, not a real equality — one is a length and one is an area.

Slide, Flip, Turn

2 MINUTES

Students translate, reflect and rotate points and shapes on a coordinate grid, work out the rule each transformation follows, and discover that all three leave every length and every angle exactly as it was.

You do not need to know this subject

Every answer is a pair of coordinates and the key prints all of them. The three rules are on the student page and nothing else is needed.

Photocopy these pages only

Pages 37–39

One per student. Do not photocopy this page or the answer key.

What students need

A pencil and a ruler. Squared paper is a bonus but the grids are printed.

WHAT TO SAY TO START

“Slide it, flip it, turn it. Those three moves are the whole of today, and they are also how every animation, every tile pattern and every game sprite on a screen actually works.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the three rules	Left is minus, down is minus
14–26	Task 7.1 — translations	Count squares, do not guess
26–38	Task 7.2 — reflections and rotations	The rules are printed
38–48	Task 7.3 — name the transformation	Work backwards
48–60	Challenge, collect sheets	Discussion question below

Questions students will ask

“Which way is positive?”

Right and up are positive. Left and down are negative. That is the same convention as every coordinate grid they have already used.

“Clockwise or anticlockwise?”

The question always says. A 180° turn is the same either way, which is why it is the easiest one on the sheet.

“Does the shape change size?”

Never, for any of these three. Lengths and angles come through untouched — only the position moves. Task 7.5 has a student who thinks otherwise.

If they finish early

Task 7.5 asks what you get if you reflect in the x-axis and then in the y-axis. It is a single rotation, and seeing that is a genuinely satisfying moment.

Curriculum focus • for the regular teacher’s records

Grade 8: describing the effect of translations, reflections and rotations on two-dimensional figures using coordinates, and understanding that these transformations preserve congruence. Discussion question for the last five minutes: *“Which of the three moves could you undo by doing it again?”*

The three moves

UNDERSTAND

16 MIN

NAME _____

CLASS _____

DATE _____

Three moves, three rules

TRANSLATE – slide

4 right, 2 down:

$$(x, y) \rightarrow (x + 4, y - 2)$$

$$(2, 3) \rightarrow (6, 1)$$

right/up +

left/down –

REFLECT – flip

in the **x-axis**:

$$(x, y) \rightarrow (x, -y)$$

in the **y-axis**:

$$(x, y) \rightarrow (-x, y)$$

$$(3, 5) \rightarrow (3, -5) \text{ or } (-3, 5)$$

ROTATE – turn

about (0, 0)

$$180^\circ (x, y) \rightarrow (-x, -y)$$

$$90^\circ \text{ anti } (x, y) \rightarrow (-y, x)$$

$$90^\circ \text{ clock } (x, y) \rightarrow (y, -x)$$

$$(4, 1) \rightarrow (-4, -1) \text{ at } 180^\circ$$

Task 7.1 • Slide it

DO

#	POINT	TRANSLATION	IMAGE
a	(2, 3)	4 right, 2 down	
b	(-1, 4)	3 left, 1 up	
c	(0, -2)	5 right, 3 up	
d	(-3, -1)	2 right, 4 up	

Task 7.2 • Flip it and turn it

DO

#	POINT	WHAT HAPPENS TO IT	IMAGE
a	(3, 5)	reflect in the x-axis	
b	(3, 5)	reflect in the y-axis	
c	(-2, 4)	reflect in the x-axis	
d	(4, 1)	rotate 180° about (0,0)	
e	(4, 1)	rotate 90° anticlockwise	
f	(-2, 3)	rotate 90° clockwise	

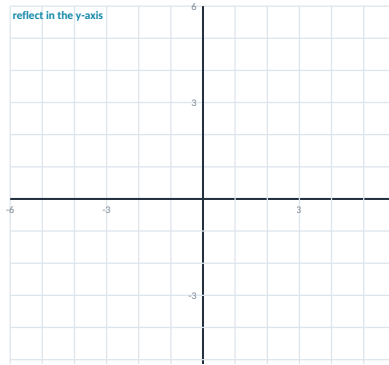
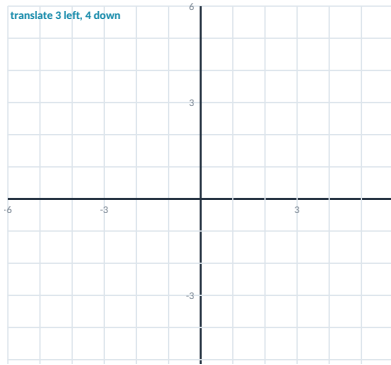
Whole shapes

22 MIN

Task 7.3 • Move the whole triangle

PLOT

Triangle **ABC** has vertices **A(1, 1)**, **B(4, 1)** and **C(1, 3)**. Plot it on the left grid, then draw and label each image.



a. $A' B' C'$ after the translation:

b. $A'' B'' C''$ after the reflection:

c. Measure AB before and after. What do you notice?

Task 7.4 • Name the transformation

REVERSE

You are given the before and after. Say which single move did it.

#	BEFORE	AFTER	WHICH TRANSFORMATION?
a	(2, 5)	(2, -5)	
b	(2, 5)	(-2, -5)	
c	(2, 5)	(7, 5)	
d	(2, 5)	(-2, 5)	
e	(2, 5)	(-5, 2)	

Undoing and combining

STRETCH

17 MIN

Task 7.5 • Find the error

DETECTIVE

Sam

"(3, 5) reflected
in the x-axis is
(-3, 5)"

What went wrong?

Correct answer: _____

Tia

"Rotating a shape
makes it smaller"

What went wrong?

Correct answer: _____

Uma

"(2, 4) moved
3 left is (5, 4)"

What went wrong?

Correct answer: _____

Task 7.6 • Think it through

CHALLENGE

- Reflect (3, 2) in the x-axis, then reflect the result in the y-axis. What single transformation would have done the same job?

- Which point does not move at all when you rotate about the origin? _____
- Which transformation can you undo by doing it a second time? _____
- A shape is translated 5 right and 2 up. Write the translation that puts it back. _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Slide, Flip, Turn

ANSWERS

Task 7.1 • Translations

- a. $(2, 3) + (4, -2) \rightarrow (6, 1)$
- b. $(-1, 4) + (-3, 1) \rightarrow (-4, 5)$
- c. $(0, -2) + (5, 3) \rightarrow (5, 1)$
- d. $(-3, -1) + (2, 4) \rightarrow (-1, 3)$

Every error here is a sign error. Left and down are negative, and a student who is counting squares on the grid rather than applying the rule rarely gets one wrong.

Task 7.2 • Reflections and rotations

- a. $(3, 5)$ x-axis $\rightarrow (3, -5)$
- b. $(3, 5)$ y-axis $\rightarrow (-3, 5)$
- c. $(-2, 4)$ x-axis $\rightarrow (-2, -4)$
- d. $(4, 1)$ 180° $\rightarrow (-4, -1)$
- e. $(4, 1)$ 90° anti $\rightarrow (-1, 4)$
- f. $(-2, 3)$ 90° clock $\rightarrow (3, 2)$

Reflecting in the x-axis changes the y, not the x, which feels backwards to most students. Say it as “the axis you flip in is the one that stays”.

Task 7.3 • The whole triangle

A(1, 1) B(4, 1) C(1, 3)

- a. translate 3 left, 4 down $\rightarrow A'(-2, -3)$ $B'(1, -3)$ $C'(-2, -1)$
- b. reflect in the y-axis $\rightarrow A''(-1, 1)$ $B''(-4, 1)$ $C''(-1, 3)$

c is the point of the task. AB is 3 units long before and 3 units long after, in both images. Every side keeps its length and every angle keeps its size — the triangles are *congruent*. That word is worth writing on the board even if the class never uses it again today.

Task 7.4 • Name the transformation

- a. $(2, 5) \rightarrow (2, -5)$ **reflection in the x-axis**
- b. $(2, 5) \rightarrow (-2, -5)$ **rotation of 180° about the origin**
- c. $(2, 5) \rightarrow (7, 5)$ **translation 5 right**
- d. $(2, 5) \rightarrow (-2, 5)$ **reflection in the y-axis**
- e. $(2, 5) \rightarrow (-5, 2)$ **rotation 90° anticlockwise**

b and e are the two students confuse. In b both signs flip and the numbers stay put; in e the numbers swap. Accept “ 270° clockwise” for e — it is the same turn.

Task 7.5 • The three errors

Sam — $(3, -5)$. Reflecting in the x-axis changes the y coordinate. He has done the y-axis instead.

Tia — no. All three transformations preserve size and shape exactly. A rotation only changes which way it faces.

Uma — $(-1, 4)$. Left means subtract. She added.

Task 7.6 • Challenge

- a. **A rotation of 180° about the origin.** $(3, 2) \rightarrow (3, -2) \rightarrow (-3, -2)$, and both signs have flipped, which is exactly what a half turn does. Two reflections in perpendicular lines always make a rotation.
- b. **The origin itself, $(0, 0)$.** It is the point being turned about.
- c. **A reflection.** Flip it and flip it again and it is back where it started. Accept a 180° rotation too, since doing that twice is a full turn.
- d. **5 left and 2 down.** Undoing a translation means reversing both directions.

Ask the Right People

READ BEFORE THE BELL

2 MINUTES

Students identify populations and samples, judge whether a sampling method is fair or biased, use a sample to estimate a whole population, and finish by designing a fair survey of their own school. The lightest lesson in the pack to supervise — most of it is reasoning, not arithmetic.

You do not need to know this subject

The only calculation is turning a fraction into a percentage and applying it to a bigger number. Everything else is judgement, and the key explains the reasoning for every answer.

Photocopy these pages only

Pages 42–44

One per student. Do not photocopy this page or the answer key.

What students need

A pencil. A calculator is optional.

WHAT TO SAY TO START

"Nobody asks all sixty-eight million people in a country who they will vote for. They ask about a thousand, and they get it close to right. Today you find out how that is possible, and what has to be true for it to work."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the sampling panel	Population, sample, bias
14–26	Task 8.1 — who is being asked?	Quick and discussable
26–38	Task 8.2 — fair or biased?	Expect argument, that is fine
38–48	Task 8.3 — estimate the population	The only arithmetic
48–55	Challenge, collect sheets	Discussion question below

Questions students will ask

"What is the difference between a population and a sample?"

The population is everybody you want to know about. The sample is the smaller group you actually ask. The sample is only useful if it looks like the population.

"What makes a sample biased?"

Anything that makes some people more likely to be picked than others. Asking outside a gym about exercise is biased before a single question is asked.

"Does a bigger sample fix bias?"

No, and this is the single most important idea in the lesson. Asking ten thousand people outside the gym gives you a very precise wrong answer.

If they finish early

Task 8.6 asks students to design a fair way to pick thirty students from seven hundred. Every sensible method involves randomness somewhere, and saying why is the real answer.

Curriculum focus • for the regular teacher's records

Grade 7: understanding that statistics can be used to gain information about a population by examining a sample, and that random sampling tends to produce representative samples. Discussion question for the last five minutes: *"Why can a survey of a thousand people predict what a whole country thinks?"*

Populations, samples and bias

UNDERSTAND

18 MIN

NAME _____

CLASS _____

DATE _____

Everybody, some of them, and what can go wrong

POPULATION

everyone you want to know about

SAMPLE

the smaller group you actually ask

BIAS

anything that makes the sample unlike the population

A FAIR SAMPLE IS RANDOM

Every member of the population must have an **equal chance** of being picked.

Names from a hat. Every tenth name on a roll.
A computer picking numbers.

FROM SAMPLE TO WHOLE

18 of 60 students walk
= 30%

School has 900 students
30% of 900 = **270**

An **estimate**, not a count.

Task 8.1 • Who is being asked?

DO

#	THE SITUATION	POPULATION	SAMPLE
a	A principal asks 40 of the school's 700 students about the canteen.		
b	A factory tests 25 of the 5,000 batteries made that day.		
c	A council surveys 200 of 12,000 residents about a new park.		

Task 8.2 • Fair or biased?

JUDGE

Write fair or biased, and if it is biased say who gets left out or over-counted.

#	THE METHOD	FAIR / BIASED	WHY
a	Asking only the basketball team about a new sports uniform.		
b	Choosing every tenth name on an alphabetical roll.		
c	Asking people outside a gym whether they exercise.		
d	Drawing 30 names from a hat holding every student.		
e	Posting an online poll and using whoever replies.		
f	Asking "Don't you agree the canteen food is terrible?"		

Estimating a whole school

KEY SKILL

20 MIN

Task 8.3 • From the sample to everybody

APPLY

Turn the sample result into a percentage, then apply it to the whole population.

#	SAMPLE RESULT	AS A PERCENTAGE	POPULATION	ESTIMATE
a	18 of 60 students walk to school		900	
b	12 of 50 name maths as their favourite		800	
c	7 of 25 batteries fail the test		5,000	
d	45 of 200 residents want the park		12,000	

Why is every one of these an estimate rather than a fact? _____

Task 8.4 • Which sample would you trust?

JUDGE

A school of 800 students wants to know how many would use a late bus. Rank these four methods from most trustworthy to least, and give a reason for your top and bottom choice.

- A. ask 12 students chosen at random
- B. ask 200 students chosen at random
- C. ask 200 students waiting at the bus stop
- D. ask all 800 students but only 60 reply

Order: _____

Best because: _____

Worst because: _____

Designing a fair survey

STRETCH

17 MIN

Task 8.5 • Find the error

DETECTIVE

Vic

"I asked my five friends, so it represents the whole school"

What went wrong?

Correct answer: _____

Wen

"20 out of 40 is 50%, so in a school of 600 it is still 20 students"

What went wrong?

Correct answer: _____

Xan

"A big enough sample is always unbiased"

What went wrong?

Correct answer: _____

Task 8.6 • Think it through

CHALLENGE

a. Describe a fair way to choose 30 students from a school of 700. Say exactly what you would do.

b. Why can a survey of about a thousand people predict a national election? _____

c. A survey outside a cinema at 10 pm on a Friday asks "how often do you go out?" Name two separate problems with it.

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Ask the Right People

ANSWERS

Task 8.1 • Who is being asked?

- a. population **all 700 students**
sample **the 40 asked**
- b. population **all 5,000 batteries**
sample **the 25 tested**
- c. population **all 12,000 residents**
sample **the 200 surveyed**

b is worth a comment: the factory cannot test all 5,000, because testing a battery to destruction leaves nothing to sell. Sampling is not always a shortcut — sometimes it is the only option there is.

Task 8.3 • Sample to population

- a. $18/60 = 30\%$ → **270 students**
- b. $12/50 = 24\%$ → **192 students**
- c. $7/25 = 28\%$ → **1,400 batteries**
- d. $45/200 = 22.5\%$ → **2,700 residents**

They are estimates because a different random sample would give a slightly different percentage. The true figure is near the estimate, not equal to it — which is why polls are reported with a margin of error.

Task 8.2 • Fair or biased?

- a. **biased** only students who already play the sport
- b. **fair** everyone has the same chance of landing on a tenth
- c. **biased** everyone there has already chosen to exercise
- d. **fair** every student is in the hat
- e. **biased** only people who feel strongly bother to reply
- f. **biased** the question tells you the answer it wants

e and f are different kinds of problem and both are worth naming. In e the *sample* is skewed because people select themselves. In f the sample might be perfect and the *question* is the problem. Students often spot the first and miss the second entirely.

Task 8.4 • Which sample would you trust?

B, then A, then D, then C is the defensible order, though B and A at the top and C at the bottom is the part that matters.

B is best — random and large. **A** is random but twelve students is too few to say much. **D** is self-selected, so the sixty who replied are probably the ones who want the bus. **C is worst** — everyone at the bus stop already catches buses, so it is biased *and* the bias points in exactly the direction being measured.

Accept D and C swapped with a good reason. The order is less important than noticing that C's two hundred people are worth less than A's twelve.

Task 8.5 • The three errors

Vic — five friends are not a sample. They are not random and people tend to be friends with people like themselves.

Wen — 300 students. She applied the sample *count* to the population instead of the sample *percentage*. 50% of 600 is 300.

Xan — no. Size and bias are separate problems. A hundred thousand people outside a gym still tells you nothing about who exercises.

Task 8.6 • Challenge

a. Anything with genuine randomness: number every student 1 to 700 and use a random number generator; put all seven hundred names in a box and draw thirty; take every twenty-third name on a full roll. What matters is that the student cannot influence who gets picked, and that no group is systematically left out.

b. Because a properly random thousand looks like the country in miniature — the same mix of ages, places and views. Accuracy comes from the randomness far more than from the size, which is why a thousand random people beat a million self-selected ones.

c. First, everybody there has already gone out, so the answer is guaranteed to be too high. Second, the time and place exclude anyone who works nights, has young children, or does not live near a cinema. Accept any two distinct problems.

What Comes Next

READ BEFORE THE BELL

2 MINUTES

Students continue number sequences, name the rule, tell arithmetic sequences from ones that multiply, and then build the n th-term rule that lets them jump straight to the fiftieth term without writing out the forty-nine before it.

You do not need to know this subject

Every rule is a small piece of arithmetic and the key gives all of them. If a student can find the gap between consecutive terms, they can do most of this sheet.

Photocopy these pages only

Pages 47–49

One per student. Do not photocopy this page or the answer key.

What students need

A pencil. A calculator is optional.

WHAT TO SAY TO START

"Here is a sequence: 3, 7, 11, 15. Anyone can give me the next one. The interesting question is the fiftieth, and by the end of the lesson you will be able to give me that in about five seconds."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the sequence panel	Find the gap first
14–26	Task 9.1 — continue and describe	Not all of them add
26–38	Task 9.2 — the n th term	The formula is built, not remembered
38–48	Task 9.3 — matchstick patterns	Draw before counting
48–55	Challenge, collect sheets	Discussion question below

Questions students will ask

"How do I find the rule?"

Work out the gap between each pair of terms. If the gap is always the same the sequence adds; if each term is a fixed multiple of the last it multiplies.

"Where does the n th term formula come from?"

The gap is the number in front of n . Then you adjust so that $n = 1$ gives the first term. Building it that way beats memorising it.

"Can a sequence have more than one valid rule?"

Yes, and Task 9.5 makes that point deliberately. Three terms are rarely enough to pin a rule down.

If they finish early

Task 9.5 gives the start 1, 2, 4 and asks for two different valid continuations. Both 8 and 7 are defensible and students enjoy arguing about it.

Curriculum focus • for the regular teacher's records

Grade 6–8: generating and analysing patterns, describing arithmetic sequences with a rule for the n th term, and distinguishing linear from non-linear growth. Discussion question for the last five minutes: *"Why are three terms usually not enough to be sure of a rule?"*

Gaps and rules

18 MIN

NAME _____

CLASS _____

DATE _____

Find the gap, then find the rule

5, 8, 11, 14 ...

gaps: +3 +3 +3

Same gap every time –
this **adds**. Next: 17, 20.

2, 6, 18, 54 ...

gaps: +4 +12 +36

Not the same. But each
term is $\times 3$ the last one.
Next: 162, 486.

THE n th TERM

5, 8, 11, 14

gap is 3, so start with **$3n$**

$n=1$ gives 3, but we want 5
so add 2 $\rightarrow$ **$3n + 2$**

term 50 = $3(50) + 2 =$ **152**

Task 9.1 • Continue it and describe it

DO

#	SEQUENCE	NEXT TWO	THE RULE IN WORDS
a	3, 7, 11, 15 ...		
b	2, 6, 18, 54 ...		
c	1, 4, 9, 16 ...		
d	100, 88, 76, 64 ...		
e	1, 1, 2, 3, 5, 8 ...		
f	5, 10, 20, 40 ...		

Only two of these add the same amount every time. Which two? _____

Task 9.2 • Build the n th term

DO

#	SEQUENCE	GAP	N TH TERM	TERM 20
a	5, 8, 11, 14 ...			
b	7, 12, 17, 22 ...			
c	20, 17, 14, 11 ...			
d	4, 10, 16, 22 ...			

Patterns you can build

KEY SKILL

20 MIN

Task 9.3 • Matchstick squares

APPLY

A row of squares is made from matchsticks. One square takes 4. Two squares in a row take 7, because they share a side. Three take 10.

squares	1	2	3	4	5	...	n
matches	4	7	10	—	—	...	—

a. Fill in the row above. _____

b. How many for 10 squares? _____

c. How many for 50 squares? _____

d. You have 100 matches. How many whole squares? _____

e. Why does the rule add 3 each time and not 4? _____

Task 9.4 • Is it in the sequence?

THINK

Use the n th term rule to decide. If n comes out as a whole number the value is in the sequence; if it does not, it is not.

#	RULE	IS THIS VALUE IN IT?	WORKING	YES / NO
a	$4n - 1$	79		
b	$3n + 1$	100		
c	$5n + 2$	60		
d	$6n - 2$	100		

More than one right answer

STRETCH

17 MIN

Task 9.5 • Find the error

DETECTIVE

Yara

"2, 4, 8, 16 adds
2, then 4, then 8,
so it is an adding
sequence"

What went wrong?

Correct answer: _____

Zed

"The n th term of
3, 6, 9, 12 is
 $n + 3$ "

What went wrong?

Correct answer: _____

Ana

"Term 10 must be
10 times term 1"

What went wrong?

Correct answer: _____

Task 9.6 • Think it through

CHALLENGE

a. A sequence starts 1, 2, 4. Give two different continuations that both follow a sensible rule, and say what each rule is.

b. Which term of $4n - 1$ is equal to 79? _____

c. Write a sequence whose terms get smaller and never reach zero. _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

What Comes Next

ANSWERS

Task 9.1 • Continue and describe

- a. 3, 7, 11, 15 → **19, 23** add 4 each time
- b. 2, 6, 18, 54 → **162, 486** multiply by 3
- c. 1, 4, 9, 16 → **25, 36** the square numbers, n^2
- d. 100, 88, 76, 64 → **52, 40** subtract 12 each time
- e. 1, 1, 2, 3, 5, 8 → **13, 21** add the two before it
- f. 5, 10, 20, 40 → **80, 160** double it

Only a and d add a constant amount. c looks like it should — but its gaps run 3, 5, 7, so it grows faster and faster. b and f multiply, and e adds a different number every time. A class that argues about c has understood the question exactly.

Task 9.2 • Build the nth term

- a. gap 3 **$3n + 2$** term 20 = **62**
- b. gap 5 **$5n + 2$** term 20 = **102**
- c. gap -3 **$23 - 3n$** term 20 = **-37**
- d. gap 6 **$6n - 2$** term 20 = **118**

c has a negative gap, so n has a minus in front of it and the sequence eventually goes below zero. Accept $-3n + 23$.

Task 9.4 • Is it in the sequence?

- a. $4n - 1 = 79 \rightarrow n = 20$ **yes**
- b. $3n + 1 = 100 \rightarrow n = 33$ **yes**
- c. $5n + 2 = 60 \rightarrow n = 11.6$ **no**
- d. $6n - 2 = 100 \rightarrow n = 17$ **yes**

c is the only one that fails, and it fails because n has to be a whole number — there is no eleven-and-a-half-th term.

Task 9.3 • Matchstick squares

squares	1	2	3	4	5	...	n
matches	4	7	10	13	16	...	$3n + 1$

- b. 10 squares → $3(10) + 1 = 31$ matches
- c. 50 squares → $3(50) + 1 = 151$ matches
- d. $3n + 1 = 100 \rightarrow 3n = 99 \rightarrow 33$ squares exactly

e is the question that matters. The first square costs four matches, but every square after it borrows one side from its neighbour, so it only costs three. That is why the rule is $3n$ plus a single extra match for the very first left-hand side — the +1 is that one match, and students who see it never forget where an n th-term constant comes from.

Task 9.5 • The three errors

Yara — it multiplies. The gaps are not equal, but each term is double the one before. A changing gap is the signal to look for multiplication.

Zed — $3n$. The gap is 3, so it starts with $3n$, and $3(1)$ is already 3, so nothing is added. Test his rule: $n = 1$ gives 4, and the sequence starts at 3.

Ana — only for sequences with no constant. In $3n$ it happens to work. In $3n + 2$, term 1 is 5 and term 10 is 32, not 50.

Task 9.6 • Challenge

a. Two good answers. Doubling gives 1, 2, 4, **8, 16**. Adding one more each time gives 1, 2, 4, **7, 11** — gaps of 1, 2, 3, 4. Both are completely valid, which is the point: three terms almost never pin a rule down, and any defended rule is correct.

b. Term 20. $4n - 1 = 79$ gives $4n = 80$.

c. Halving works: 100, 50, 25, 12.5 ... It gets closer to zero forever without arriving. Accept any sequence that shrinks by a fraction rather than by a fixed amount.

Two Ways to Count

READ BEFORE THE BELL

2 MINUTES

Students complete two-way tables from partial information, read percentages out of them, build a table from a written description, and finish by deciding whether two things are actually associated or just look that way.

You do not need to know this subject

Every cell is found by subtracting from a total, and every total must agree both across and down. That built-in check means a student can tell whether their own table is right before you do.

Photocopy these pages only**Pages 52–54**

One per student. Do not photocopy this page or the answer key.

What students need

A pencil. A calculator helps with the percentages.

WHAT TO SAY TO START

"Eighty per cent of pet owners live in a house. Sixty per cent of everyone lives in a house. Is owning a pet connected to living in a house, or is that just what the numbers happen to look like? Today you learn to tell the difference."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the table panel	Totals across and down must agree
14–26	Task 10.1 — complete the table	Subtraction, nothing harder
26–38	Task 10.2 — reading it	Careful which total is the denominator
38–50	Task 10.3 — build one from words	Slowest part, expect it
50–55	Challenge, collect sheets	Discussion question below

Questions students will ask**"Which number do I divide by?"**

Whichever group the question is about. "What fraction of Year 8 walk" divides by the Year 8 total; "what fraction of walkers are Year 8" divides by the walking total. Same cell, two different answers.

"How do I know my table is right?"

Add the row totals and add the column totals. Both must give the same grand total. If they do not, something is wrong and you can find it yourself.

"What does association mean?"

That knowing one thing changes the odds on the other. If 80% of pet owners live in houses but only 55% of non-owners do, the two are associated.

If they finish early

Task 10.6 asks whether association proves that one thing causes the other. It does not, and coming up with a third explanation is the best five minutes in the lesson.

Curriculum focus • for the regular teacher's records

Grade 8: constructing and interpreting two-way frequency tables, using relative frequencies to describe possible association between two variables.

Discussion question for the last five minutes: *"If two things go together, does one of them have to be causing the other?"*

Reading a two-way table

UNDERSTAND

18 MIN

NAME _____

CLASS _____

DATE _____

Every number has to agree with its neighbours

	walk	bus	total
Year 7	24	36	60
Year 8	18	42	60
total	42	78	120

Rows add across.
Columns add down.
Both reach 120.

WHICH TOTAL?

"What fraction of Year 8 walk?"

$$\rightarrow 18 \div 60 = 30\%$$

"What fraction of walkers are Year 8?"

$$\rightarrow 18 \div 42 = 43\%$$

Same cell. Different question.

FILLING GAPS

If you know the row total and one cell, subtract.

$$60 - 24 = 36$$

Start wherever you have two out of three.

Task 10.1 • Complete the table

DO

A school surveyed 120 students about how they get to school.

	WALK	BUS	TOTAL
Year 7	24		60
Year 8		42	
Total			120

Check it: do your row totals and your column totals both reach 120? _____

Task 10.2 • Read it carefully

DO

#	QUESTION	WORKING	ANSWER
a	How many Year 7 students walk?		
b	What percentage of Year 8 walk?		
c	What percentage of all students walk?		
d	Of the students who walk, what percentage are Year 7?		

Parts **b** and **d** both involve walking and a year level, and the answers are different. Why? _____

Task 10.3 • Build one from words

APPLY

Two hundred people were surveyed. **120 own a pet**. Of the pet owners, **96 live in a house**. Altogether **140 people live in a house**. Fill in every cell.

	HOUSE	FLAT	TOTAL
Owns a pet			
No pet			
Total			

Which three numbers did you write in first, and why those? _____

Task 10.4 • Is there an association?

THINK

Use your completed table from Task 10.3.

a. What percentage of *pet owners* live in a house?

c. What percentage of *everyone* lives in a house?

b. What percentage of *non-owners* live in a house?

d. Are pet ownership and housing associated? _____

e. If there were *no* association at all, what would you expect a and b to look like? _____

Association is not cause

STRETCH

17 MIN

Task 10.5 • Find the error

DETECTIVE

Bo

"96 of 200 people are pet owners in houses, so 96% of pet owners are"

What went wrong?

Correct answer: _____

Cy

"My row totals give 200 so the table must be right"

What went wrong?

Correct answer: _____

Di

"Pets and houses are associated, so pets make people buy houses"

What went wrong?

Correct answer: _____

Task 10.6 • Think it through

CHALLENGE

- a. Ice cream sales and drowning both rise in the same months. Suggest a third thing that explains both.

- b. In a table of 200 people, 100 are in each row and 100 in each column. What must every cell be if there is no association at all?

- c. Why is a two-way table more useful than two separate lists? _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Two Ways to Count

ANSWERS

Task 10.1 • The completed table

	walk	bus	total
Year 7	24	36	60
Year 8	18	42	60
total	42	78	120

Start with $60 - 24 = 36$, then $120 - 60 = 60$ for the Year 8 total, then $60 - 42 = 18$. Both the row totals and the column totals reach 120, which is the check.

Task 10.2 • Reading it

- a. **24 students**
- b. $18 \div 60 = 30\%$
- c. $42 \div 120 = 35\%$
- d. $24 \div 42 = 57\%$ (to the nearest per cent)

b and d differ because the denominators differ. b asks about Year 8 students, so it divides by 60. d asks about walkers, so it divides by 42. Getting the denominator from the wording is the whole skill in this lesson.

Task 10.3 • Built from words

	House	Flat	Total
Owns a pet	96	24	120
No pet	44	36	80
Total	140	60	200

The three given numbers are 120, 96 and 140. From those: $120 - 96 = 24$ flats with pets; $200 - 120 = 80$ without pets; $140 - 96 = 44$ non-owners in houses; then $80 - 44 = 36$, and the flat column checks out at $24 + 36 = 60$. A student who works outward from the two totals they were handed will never get stuck.

Task 10.4 • Is there an association?

- a. $96 \div 120 = 80\%$ of pet owners
- b. $44 \div 80 = 55\%$ of non-owners
- c. $140 \div 200 = 70\%$ of everyone
- d. **yes — clearly**

Eighty per cent against fifty-five is a large gap, so knowing whether somebody owns a pet really does change your estimate of where they live. **e:** with no association at all, a and b would both be about 70% — the same as everyone else — because knowing about the pet would tell you nothing new.

Task 10.5 • The three errors

Bo — 80%. He divided by the grand total instead of by the number of pet owners. 96 out of 120, not out of 200.

Cy — not enough. The rows agreeing proves nothing on its own; the columns have to reach 200 as well. Both checks or neither.

Di — association is not cause. People who already have space and a garden are more likely to get a pet, which is the arrow pointing the other way entirely.

Task 10.6 • Challenge

a. Hot weather. It drives ice cream sales up and it also sends people swimming, which is where drownings happen. Neither one causes the other; a third thing causes both. Accept summer, school holidays, or anything with the same shape.

b. 50 in every cell. If the two things are unrelated, each row splits in the same proportion as the whole table — and here that is half and half.

c. Because two separate lists tell you how many own pets and how many live in houses, but never how those overlap. The table is the only one of the three that can show you an association, and that is exactly what it is for.

Handover note

Thank you for covering this class. Please spend two minutes on this page before you leave and put it on the desk — it is the only handover the regular teacher will get.

SUBSTITUTE'S NAME

DATE

Class / period

Room

WHICH LESSON DID YOU USE?

- | | |
|---|--|
| ☐ Lesson 01 • Rounding & estimation | ☐ Lesson 02 • Decimal $\times$ and $\div$ |
| ☐ Lesson 03 • Measurement & conversion | ☐ Lesson 04 • Absolute value & opposites |
| ☐ Lesson 05 • Variables on both sides | ☐ Lesson 06 • Circumference & area of circles |
| ☐ Lesson 07 • Transformations | ☐ Lesson 08 • Sampling, bias & inference |
| ☐ Lesson 09 • Sequences & patterns | ☐ Lesson 10 • Two-way tables & frequency |

How far did the class get?

- ☐ Finished everything, including the challenge
- ☐ Finished the first two pages
- ☐ Finished the first page only
- ☐ Did not get started — see notes below

The sheets

- ☐ Collected and left on the desk
- ☐ Students kept them
- ☐ Marked together in class using the key

ANYTHING THE TEACHER SHOULD KNOW

STUDENTS WHO WORKED ESPECIALLY WELL

STUDENTS WHO NEEDED MORE SUPPORT

From one teacher to another

Covering a class you do not teach, for students you do not know, from a folder you have never seen is genuinely hard. If today was messy, that is not a reflection on you. Write down what happened, leave the page here, and go home. Thank you.

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Standards alignment

Written against the Grade 6–8 band of the Common Core State Standards for Mathematics, and readily mapped to state equivalents and to the Australian Curriculum Years 7–8.

- **01 • Close Enough** — Rounding & estimation
- **02 • Point of Order** — Decimal $\times$ and $\div$
- **03 • Change of Units** — Measurement & conversion
- **04 • Both Directions** — Absolute value & opposites
- **05 • Balancing Act** — Variables on both sides
- **06 • Round Numbers** — Circumference & area of circles
- **07 • Slide, Flip, Turn** — Transformations
- **08 • Ask the Right People** — Sampling, bias & inference
- **09 • What Comes Next** — Sequences & patterns
- **10 • Two Ways to Count** — Two-way tables & frequency

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